



Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division

Digital Learning Handout

Part A: Content Design

Course Title	Essential Mathematics for Data Science
Course No(s)	DSE*ZC411
Credit Units	4
Credit Model	4-0-0 (Lectures units - Tutorial units - Practical (Labs) units)
Instructors	Deepak G Skariah
Version No:	1
Date:	09.06.2026

Course Description:

Linear algebra, vectors and matrices, matrix operations, vector spaces and linear transformations, eigenvalues and eigenvectors, matrix factorizations (LU, QR, SVD), norms and distance measures, multivariate calculus, limits and continuity, differentiation and partial derivatives, gradients and Jacobians, optimization fundamentals, convex sets and convex functions, unconstrained optimization, probability theory, random variables and probability distributions, expectation and variance, joint and conditional distributions, Bayes theorem, statistical inference, sampling methods, confidence intervals, hypothesis testing, correlation and covariance, regression fundamentals, multivariate statistics, numerical methods for data science.

Course Objectives

No	Course Objective
C01	Linear algebra foundations - To introduce students to the fundamental concepts of linear algebra, including vectors, matrices, vector spaces, and linear transformations, as tools for representing and manipulating data in high-dimensional spaces.
C02	Matrix structure and decomposition techniques - To develop an understanding of matrix operations and decompositions such as LU, QR, and SVD, and to highlight their role in solving linear systems.
C03	Multivariate Calculus and data modeling - To provide foundational knowledge of multivariate calculus, including limits, continuity, partial derivatives, gradients, Jacobians, and Hessians, with applications to data-driven modeling and optimization.
C04	Optimization principles in data science – To introduce the principles of optimization, including convex sets and convex functions, and to develop intuition for unconstrained optimization methods used in machine learning algorithms.
C05	Probability theory and stochastic modeling – To build a strong foundation in probability theory, including random variables, probability distributions, expectation, variance, and joint and conditional distributions for modeling uncertainty in data.





CO6	Statistical inference and data interpretation - To introduce statistical inference techniques such as sampling methods, estimation, confidence intervals, and hypothesis testing for drawing conclusions from the data.
CO7	Numerical and statistical methods for computational data science – To expose students to numerical methods and multivariate statistical techniques, including regression, covariance analysis, and computational approaches for solving real-world data science problems.

Text Book(s):

T1	M.P. Dienesroth, A. Aldo Faisal, Cheng Soon Ong, Mathematics for Machine Learning, Cambridge University Press, 2020.
T2	G. Strang, Introduction to Linear Algebra, 4 th Edition, Wellesley-Cambridge Press, 2009.
T3	L. Wasserman, All of Statistics: A Concise Course in Statistical Inference, Springer 2004.

Reference Book(s) & other resources:

R1	K Hoffman and R Kunze, Linear Algebra, Pearson Education, 2 nd Edition, 2005.
R2	Erwin Kreyszig, Advanced Engineering Mathematics, Wiley India, 10th Edition, 2015 (earlier editions are also okay)

Learning Outcomes: Students will be able to

LO1	Students will be able to represent and manipulate data using vectors, matrices, and linear transformations to model real-world data science problems.
LO2	Students will be able to compute and interpret eigenvalues, eigenvectors, and matrix factorizations such as LU, QR, and SVD for analyzing data structure and dimensionality reduction.
LO3	Students will be able to apply multivariate calculus concepts including partial derivatives, gradients, Jacobians, and Hessians to analyze and optimize multivariable functions.
LO4	Students will be able to formulate and solve unconstrained optimization problems using gradient-based and convex optimization techniques relevant to machine learning models.
LO5	Students will be able to apply probability theory, random variables, and joint distributions to model uncertainty and compute expectations, variances, and conditional probabilities.
LO6	Students will be able to perform statistical inference using sampling distributions, confidence intervals, and hypothesis testing to draw valid conclusions from data.
LO7	Students will be able to implement numerical and statistical methods such as regression, covariance analysis, and simulation-based techniques to solve computational data science problems.

Modular Content Structure:

1. Basic vector algebra

- Vectors and matrices
- matrix operations





- dot product
- geometric interpretation
- systems of linear equations.

2. Vector spaces, linear independence

- Vector spaces
- Subspaces
- Span
- linear independence
- basis and dimension
- linear independence
- linear transformations.

3. Matrix decomposition

- Determinant and trace
- Eigenvalues and eigenvectors
- LU factorization
- QR factorization
- Singular-value decomposition
- Low-rank approximation

4. Multivariate Calculus

- limits and continuity
- differentiation and partial derivatives
- chain rule
- gradients
- Jacobians

5. Optimization

- convex functions and convex sets
- gradient descent
- unconstrained optimization
- learning rates
- regularization

6. Probability

- probability theory
- random variables and probability distributions.
- expectation and variance
- joint and conditional distributions



7. Statistical Inference

- sampling methods
- confidence intervals
- hypothesis testing
- correlation and covariance
- regression fundamentals
- multivariate statistics

8. Numerical methods for data science

- numerical conditioning
- floating-point errors
- iterative solvers
- numerical stability
- condition number

Part B: Learning Plan

Contact Session	List of Topic Title	Sub-Topics	Reference
1	Vectors, matrices and systems of linear equations	• Vectors and matrices • Matrix operations • Dot product • Geometric interpretation • Systems of linear equations	T1: Sec 2.1, 2.2, 2.3. T2: Sec 1.1 – 1.4
2	Vectors spaces and subspaces	• Vector spaces and subspaces • Span of vectors • Linear independence	T1: Sec 2.4, 2.5, 2.6, T2: Chapter 3
3	Vector spaces and	• Linear independence	T1: Sec 2.4, 2.5, 2.6,





	linear transformations	• Basis and dimension • Linear transformations	T2: Chapter 3
4	Matrix decomposition	• Eigenvalues and eigenvectors • Diagonalization • Symmetric matrices • Covariance matrices • PCA intuition	T1: Sec 4.4, 4.5, 4.6 T2: Chapters 5-6
5	Matrix factorization	• LU factorization • QR factorization • Singular-value decomposition • Low-rank approximation	T1: Sec 4.4, 4.5, 4.6
6	Multivariate Calculus	• Limits and continuity • Differentiation and partial derivatives • Chain rule	T1: Sec 5.1, 5.2, 5.3
7	Gradients	• Gradients • Jacobians • Hessians • Taylor approximation • Optimization fundamentals	T1: Sec 5.5, 5.6
8	Optimization fundamentals	• Convex functions and convex sets • Gradient descent • Unconstrained optimization • Learning rates • regularization	T1: Sec 7.1, 7.3
9	Probability theory	• probability spaces • conditional probability • Bayes theorem • independence	T3: Chapter 1
10	Random variables	• Discrete and continuous random variables • Expectation and variance • Joint and conditional distributions • Common distributions	T3: Chapter 2
11	Joint distributions	• Joint distributions • Conditional distributions • Covariance matrices • Correlation	T3: Chapter 2, 3
12	Sampling and limit theorems	• Law of large numbers • Central limit theorem • Sampling distributions • Monte-Carlo intuition	T3: Chapter 3, T3: Chapter 5
13	Statistical inference	• Point estimation	T3: Chapter 6





		• Maximum likelihood • Confidence intervals	
14	Hypothesis testing and regression	• Null hypothesis • P-values • Linear regression • Least squares	T3: Chapter 6, Chapter 10, Chapter 13
15	Multivariate statistics	• Multivariate Gaussian distribution • Covariance geometry • Clustering intuition • Gaussian mixture models	T1: Chapter 6, T3: Chapter 13, Chapter 14
16	Numerical methods in data science	• Numerical conditioning • Floating point errors • Iterative solvers • Numerical stability • Condition number	T2: Chapter 9, Class Notes

Experiential Learning Components:

Describe objective, outcome of Experiential Learning Component and the lab infrastructure needed (virtual, remote, open source etc..) number of lab exercises needed, etc.

1. Lab work: 8
2. Project work: none
3. Case Study: none
4. Simulation: none
5. Work Integrated Learning Assignment: none
6. Design work/ Field work: none

Objective of Experiential Learning Component:

Hands on sessions on the following:

- a) Vectors and Matrices: create vectors and matrices in NumPy/Octave and solve for $Ax=b$.





- b) Geometry of vector spaces – compute projections, construct different bases.
- c) Eigenvalues, eigenvectors and Principal Components Analysis – compute eigenvalues and eigenvectors, perform PCA manually and compare with SciKit-Learn PCA.
- d) Compute SVD and low-rank approximation of a matrix. Observe compression effects on an image.
- e) Multivariate Calculus and optimization – numerically approximate derivatives and compare with analytical derivatives.
- f) Gradient descent – implement gradient descent for simple functions.
- g) Probability and Monte-Carlo simulation.
- h) Bayesian reasoning

Confidence intervals and hypothesis testing

Scope of Experiential Learning Component:

Programming language: Python/C/Octave

Tools and libraries: Jupyter, Numpy, Scipy, Pandas, ScikitLearn, Matplotlib, Seaborn, PGMPY

Lab Infrastructure:

Online / Open source / Virtual lab / Google Colab

List of Experiments:

Exp No.	Experiment Title	Reference to handout module/section
1	Solving $Ax = b$ using Octave and studying the relationship of the accuracy of the solution with the condition number of the matrix.	1,2
2	Geometry of vector spaces – compute projections, construct different bases for a vector space.	2,3
3	Compute eigenvalues and eigenvectors, perform PCA with Scikit-Learn/Octave.	4,5
4	Compute SVD and low-rank approximation of a matrix. Observe compression effects on an image.	4,5
5	Compute derivatives numerically and compare with analytical derivatives.	5





6	Implement gradient descent for simple functions such as $f(x) = x^2$.	6,7,8
7	Find the area of a geometric object using Monte-Carlo simulation. Study the effect of increasing the number of trials and compare with the true value.	9,10,11, 12
8	Bayesian reasoning – construct a spam filtering example, compute priors, likelihoods and posteriors.	12, 13
9	Confidence intervals and hypothesis testing – using real datasets, estimate mean, and construct 95%, 99% confidence intervals. Perform one-sample t-test, two-sample t-test.	14,15

Evaluation Scheme:

Legend: EC = Evaluation Component; AN = After Noon Session; FN = Fore Noon Session

Evaluation Component	Name (Quiz, Lab, Project, Mid-term exam, End semester exam, etc.)	Type (Open book, Closed book, Online, etc.)	Weight	Duration	Day, Date, Session, Time
EC – 1*	Quiz	Online	10%	1 week	August 10-20, 2026
	Assignment/Lab Assignment / Lab Exams	Online	20 %	10 days	August 27-7 Sept 2026
EC - 2	Mid-Semester Test	Closed Book	30%	2 hours	19/09/2026 (FN)
EC - 3	Comprehensive Exam	Open Book	40%	2 ½ Hours	05/12/2026 (FN)

EC1* (20% - 30%): Quiz (optional): 5-10 %, Lab Assignment/Assignment: 20% - 30%

Syllabus for Mid-Semester Test (Closed Book): Topics in Contact session: 1 to 8

Syllabus for Comprehensive Exam (Open Book): All topics

Important Links and Information:



eLearn Portal: <https://elearn.bits-pilani.ac.in>

Students must visit the eLearn portal regularly and stay updated with the latest announcements and deadlines.

Contact Sessions: Students should attend the online lectures as per the schedule provided on the eLearn portal.

Evaluation Guidelines:

1. EC-1 consists of either two Assignments or three Quizzes. Students will attempt them through the course pages on the eLearn portal. Announcements will be made on the portal in a timely manner.
2. For Closed Book tests: No books or reference material of any kind will be permitted.
3. For Open Book exams: “open book” means text/ reference books (publisher copy only) and does not include any other learning material. No other learning material will be permitted during the open book examinations. For Detailed Guidelines refer to the attached document.

EC3 Guidelines

4. If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student should follow the procedure to apply for the Make-Up Test/Exam, which will be made available on the eLearn portal. The Make-Up Test/Exam will be conducted only at selected exam centres on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course handout, attend the online lectures, and take all the prescribed evaluation components such as Assignments/Quizzes, Mid-Semester Tests and Comprehensive Exams according to the evaluation scheme provided in the handout.

